KEVIN MWEMA MUTUA

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PROFESSIONAL SUMMARY

Solutions-oriented Electronics and Computer Engineering student at JKUAT with hands-on experience in Data Science, Machine Learning, and Cybersecurity. Proven track record of developing machine learning classifiers, NLP threat detection models, and software automation pipelines. Seeking an engineering attachment/internship to leverage technical expertise in data science, AI application development, and security engineering.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, C, Kotlin, SQL
- **Data Science & ML:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Feature Engineering, EDA
- **AI & Security:** NLP (TF-IDF), Random Forest Classifiers, Malware Analysis Concepts, Network Security (TCP/IP)
- **Tools & Environments:** Git/GitHub, VS Code, Jupyter Notebook, Linux, Windows, Arduino IDE, Gradio, Hugging Face

FEATURED PROJECTS

Human vs Machine – AI Text Classifier

- **Situation/Task:** Standard AI detection tools often produce high false-positive rates or overfit due to flawed baseline numerical features in datasets.
- **Action:** Engineered a custom text-classification pipeline recalculated directly from raw text using 7 linguistic metrics (Flesch Reading Ease, Gunning Fog Index, sentence variance) and trained a Random Forest model using 5-fold cross-validation and GridSearchCV.
- **Result:** Successfully built a transparent, reproducible classifier integrated with a Gradio web dashboard for real-time human vs. AI text detection.

Social Sentinel – AI-Powered Phishing & Social Engineering Analyzer

- **Situation/Task:** Organizations require fast, automated detection of psychological manipulation and threat vectors in incoming communication.

- **Action:** Developed an open-source NLP analyzer using TF-IDF vectorization (unigrams/bigrams) and evaluated text across 4 classification models (Naive Bayes, Logistic Regression, Random Forest, SVM).
- **Result:** Implemented real-time detection of urgency, authority exploitation, and psychological manipulation with interactive confidence scores on Gradio and Hugging Face.

End-to-End Marketing & Customer Pattern Analytics

- **Situation/Task:** Raw marketing datasets required structured statistical inference to identify actionable customer targeting segments.
- **Action:** Conducted end-to-end data cleaning, feature engineering, and statistical hypothesis testing in Python using Pandas and Seaborn.
- **Result:** Identified key customer behavioral patterns and generated data-driven recommendations to optimize marketing campaign performance.

EXPERIENCE

Data Science & Machine Learning Attaché | Wecncode

Summer 2026

- **Action:** Applied advanced Python data processing, feature engineering, and statistical analysis across multiple capstone projects.
- **Result:** Completed capstone projects in statistical inference and customer pattern analysis, improving dataset visualization clarity and model predictive power.

Electrical Engineering Attaché | Kenya Airports Authority (KAA), Wilson Airport

Summer 2025

- **Action:** Executed routine preventive maintenance, troubleshooting, and inspections on critical airfield electrical infrastructure, including runway lighting systems and back-up power generators.
- **Result:** Ensured continuous operational uptime and compliance with aviation safety standards across airport facility systems.

EDUCATION & CERTIFICATIONS

Jomo Kenyatta University of Agriculture and Technology (JKUAT)

B.S. Electronics and Computer Engineering (Expected May 2028)

- **Relevant Coursework:** Data Structures & Algorithms, Data Science, Operating Systems, Computer Networks, Circuit Theory.

Certifications

- **IBM Data Science Professional Certificate** – IBM
- **Cybersecurity Fundamentals (Capstone Project)** – Credly / IBM
- **Android Application Development** – Zalego Academy

LEADERSHIP & ACTIVITIES

- **Laboratory Captain (Utumishi Academy):** Enforced laboratory safety protocols and organized equipment for over 100+ practical science sessions.
- **University Football Team:** Demonstrated discipline, time management, and collaborative teamwork in competitive university athletics.